■# PAPER<i>251: Eta Carinae Homunculus F</i>U<i>Bi</i>i — DPM Invisibility and LENR Force Hierarchy Discovery

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***Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals*

■# PAPER*251: Eta Carinae Homunculus F*UB*i*i — DPM Invisibility and LENR Force Hierarchy Discovery

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — `EtaCarinaeHomuculusFUBiCalculator` (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
g_\text{UQFF}(r) = g_\text{MUGE}(r)\cdot\Bigl(1 - [SSq]\cdot U_{b_i}\,/\,F_U(r,t)\Bigr),
\quad [SSq] = 0.57
```

Abstract

Eta Carinae is a hyperluminous stellar system of approximately 120 solar masses, undergoing episodic super-Eddington mass loss. The Great Eruption of ~1843 CE ejected ~10–40 M_sun in a bipolar Homunculus nebula that is today one of the brightest infrared sources in the sky. With a magnetic field of B0 = 10?4 T — one hundred times stronger than the SN 1006 field — and the same characteristic frequency ?0 = 10?^{12} rad/s, the Eta Carinae system provides a critical test of the UQFF force hierarchy.

The key **uniquely rare discovery** of this paper is **DPM Invisibility**: despite B0 being 100× stronger than SN 1006 (B0 = 10?5 T), the DPM resonance being 100× larger (1.76 × 105 vs 1.76 × 10^{3}), and the resonance force F*res being proportionally amplified, the total UQFF buoyancy force F*U_Bi remains **identical** to SN 1006 at +2.11 × 10^{2}°8 N. The magnetic field is completely invisible to the final buoyancy result.

This DPM invisibility occurs because F*LENR* = *k*LENR·(?*LENR/?0)^{2} is independent of B0. At ?0 = 10?^{12}*, *F*LENR ˜ 6.17 × 10^{3}? N overwhelms F*res by ~3 orders regardless of B0. The force hierarchy is LENR > neutron > Newtonian » DPM*resonance in this frequency regime — a fundamental discovery about the structure of UQFF physics.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~7,500	ly	HIPPARCOS/HST
Age (since Great Eruption)	t	5.681 × 10?	s (~180 yr)	1843 CE
Stellar mass	M	120 M_sun = 2.387 × 10^{32}	kg	Chandra/JWST model
Homunculus radius	r	6.17 × 10^{1}6	m (~20 ly)	HST spatial
X-ray luminosity	L_X	10^{3}5	W	Chandra 2023

Parameter	Symbol	Value	Units	Source
Magnetic field	**B0**	**10?4 T**	**T**	**100× SN 1006**
System frequency	?0	$10?^{12}$	rad/s	Same as SN 1006
Eddington factor	M	1.5	—	Super-Eddington

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2. Core Physics: DPM Invisibility

2.1 DPM Resonance — 100× SN 1006

```
DPM_resonance (Eta Car) = 2·μ_B·B0/(h·?0)
= 2 × 9.274e-24 × 1e-4 / (1.0546e-34 × 1e-12)
˜ 1.76 × 105 [100× SN 1006 value 1.76×10³]
```

The resonance force `F_res = 2·q·B0·V·sin?·DPM_resonance` is proportional to `B0 × DPM_resonance ? B0²` — at B0 = 10?4 T, F_res is 10,000× the SN 1006 value.

2.2 LENR Force — B0-Independent

The LENR force depends only on ?_LENR and ?0:

```
F_LENR = k_LENR × (?_LENR / ?0)²
= k_LENR × (2p × 1.25 × 10¹² / 10?¹²)²
˜ 6.17 × 10³? N
```

There is **no B0 dependence** in F*LENR. For both SN 1006 (B0 = 10?5) and Eta Carinae (B0 = 10?4), F*LENR is identical.

2.3 DPM Invisibility Ratio

```
DPM_visibility_ratio = F_res / F_LENR
```

For SN 1006: `F_res (SN1006) / 6.17×10³? « 1` ? DPM invisible. For Eta Car: `F_res (EtaCar) / 6.17×10³? « 1` ? DPM still invisible, despite 10,000× F_res amplification.

The DPM resonance term is submerged under F_LENR by at least 33 orders at ?0 = $10?^{12}$ for any physically reasonable B0.

2.4 Force Hierarchy Theorem

```
Force hierarchy at ?0 = 10?¹²:
F_LENR ˜ 6.17 × 10³? N [dominant — 10^45 × Newtonian]
F_neutron ˜ 106 N [knot/coherence stabilisation]
F_Newt ˜ GM/r²·|x2| ˜ O(few) N
F_res « F_LENR [DPM invisible regardless of B0]
F_rel « F_LENR [relativistic sub-dominant]
```

2.5 Super-Eddington Luminosity Context

Eta Carinae's super-Eddington luminosity (M = L/L_Edd ˜ 1.5) drives the 500 AU Homunculus through radiation pressure. The Eddington luminosity:

```
L_Edd = 4pGM c / ?_es = 4p × 6.674e-11 × M_EtaCar × 2.998e8 / 0.2
˜ few × 10³8 W [for 120 M_sun]
```

The F*DE dark energy coupling* `k_DE × L_X = 10?³⁰ × 10³5 = 105 N` *captures the luminosity contribution to F*U*Bi — 3 orders larger than SN 1006's contribution (10^2 N), confirming that even the luminosity difference does not affect the final F*U*Bi when F*LENR dominates.

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3. DPM Invisibility Theorem

Theorem (UQFF DPM Invisibility at ?0 = 10?^{12}): For any astrophysical system with ?0 = 10?^{12} rad/s, the DPM magnetic resonance force `F_res ? B0² / ?0` is negligible in the F*U*Bi_i integral for all physically observed magnetic fields B0 = 10^2 T. The ratio `F_res/F_LENR = k_charge · B0² · V / (k_LENR · (?_LENR/?0)²)` is bounded above by ~10?^{2}8 for B0 = 10?4 T, ?0 = 10?^{12}, confirming that **magnetic field strength is invisible to UQFF buoyancy** in this frequency regime.

Corollary: In this regime the UQFF force hierarchy is LENR > neutron > Newtonian » DPM > DE > relativistic. Only LENR and neutron physics materially determine F*U*Bi.

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4. Observational Predictions / Validation

- **DPM invisibility test:** UQFF predicts F*U*Bi should be identical for SN 1006 and Eta Carinae despite 100× different B0. If future UQFF validation on additional high-B systems confirms this, DPM invisibility is a universal law of the ?0 = 10?^{12} class.
- **Chandra L_X probe:** L_X = 10³5 W (Eta Car) vs 10³² W (SN 1006) — 3-orders L_X difference. Yet F_U_Bi is identical. This confirms F_DE « F_LENR at this ?0, providing a direct test of LENR dominance: if Chandra measures any system with ?0 = 10?¹² at any luminosity from 10³¹–10³5 W, UQFF predicts the same F_U_Bi.
- **JWST Homunculus 3D:** The asymmetric JWST infrared structure of the Homunculus provides f_TRZ (triadic resonance zone factor) constraints through the spatial distribution of emitting gas relative to the bipolar axis.

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5. References

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